

INTERNSHIP PROJECT REPORT

ON

AADHAAR ENROLLMENT ANALYSIS

Unlocking Societal Trends in Aadhaar Enrollment and Updates

Submitted in partial fulfilment of the requirements of the
Internship Program at

APPAXON SOLUTIONS

Submitted by
Vanshika Saini

Project Manager: Vivek Jha

Appaxon Solutions
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Certificate

This is to certify that Vanshika Saini has successfully completed the internship project titled “Aadhaar Enrollment Analysis” at Appaxon Solutions under the supervision and guidance of Mr. Vivek Jha, Project Manager.

During the course of this internship, the intern worked on analyzing Aadhaar enrollment data for the state of Maharashtra, applying data cleaning, feature engineering, and exploratory data analysis techniques to derive meaningful insights. The work carried out reflects a sincere effort, technical understanding, and a methodical approach towards problem-solving.

We wish her success in all future endeavors.

Vivek Jha

Project Manager, Appaxon Solutions

Acknowledgement

I would like to express my sincere gratitude to Appaxon Solutions for providing me the opportunity to work on this internship project. The exposure to real-world data analysis has been a valuable learning experience for me.

I am especially thankful to Mr. Vivek Jha, my Project Manager, for his constant guidance, patience, and support throughout the duration of this project. His feedback at every stage helped me approach the problem in a structured manner and improve the quality of my work.

I would also like to thank my institution for equipping me with the foundational knowledge in data analysis that made this project possible. Finally, I extend my thanks to everyone who directly or indirectly supported me during this internship.

Vanshika Saini

Abstract

Aadhaar is the primary identification system used across India, and tracking its enrollment activity helps understand how identity coverage is progressing across regions and age groups. This project analyzes Aadhaar enrollment data for the state of Maharashtra covering the period from January 2025 to July 2026, consisting of 537,489 enrollment records across 36 districts and 1,604 pincodes.

The dataset was cleaned and enriched with derived time-based features such as year, month, quarter, and year-month, along with a computed total enrollment field. Exploratory data analysis was carried out to study monthly and quarterly enrollment trends, district-wise and pincode-wise distribution, age-group-wise participation, and month-over-month growth behaviour. A statistical z-score method was also used to flag months with unusually high enrollment activity.

The analysis shows that enrollment is concentrated in the 0–5 years age group, that a small set of districts and pincodes account for a disproportionately large share of total enrollment, and that enrollment activity peaked sharply in February and July of 2025. These insights can support Aadhaar-issuing authorities in planning resource allocation, staffing, and awareness campaigns across the state.

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1. Introduction

Aadhaar is a twelve-digit unique identity number issued to residents of India, used as proof of identity and address across government and private services. The scale at which Aadhaar enrollment happens across a state as large as Maharashtra generates a continuous stream of data that can reveal patterns in how, where, and when people are enrolling.

This report documents the work carried out during the internship at Appaxon Solutions on the project “Aadhaar Enrollment Analysis.” The project involved working with a raw enrollment dataset for Maharashtra, cleaning it, engineering useful time-based features, and performing exploratory data analysis to identify enrollment trends across time, geography, and age groups.

The internship provided practical exposure to a complete data analysis workflow — from loading raw data in Python, through cleaning and transformation, to visual analysis and interpretation of results — using libraries such as Pandas, NumPy, Matplotlib, Seaborn, and SciPy.

2. Problem Statement

Raw Aadhaar enrollment records, by themselves, do not reveal any pattern or trend. Without structured analysis, it is difficult to know which districts or pincodes have high enrollment activity, which time periods see enrollment spikes, or which age groups are under-enrolled.

The problem addressed in this project is to convert the raw enrollment dataset for Maharashtra into meaningful, structured insights covering temporal patterns, geographical distribution, and age-group participation, so that these insights can support planning and decision-making around Aadhaar enrollment services.

3. Objectives

- To clean and prepare the Aadhaar enrollment dataset for analysis.
- To engineer time-based features (year, month, quarter, day) from the enrollment date.
- To identify monthly and quarterly enrollment trends across the study period.
- To analyze district-wise and pincode-wise variation in enrollment activity.
- To study enrollment distribution across age groups (0–5, 5–17, 18+ years).
- To detect anomalous months using statistical methods.
- To translate the findings into insights and recommendations useful for planning.

4. Scope of the Project

This study analyzes Aadhaar enrollment data for Maharashtra from January 2025 to July 2026. The analysis focuses on identifying temporal trends, geographical distribution across districts and pincodes, and age-group-wise enrollment patterns.

The findings are limited to the available enrollment dataset and do not include demographic updates, biometric updates, or comparisons with other states. The 2026-Q3 quarter in the dataset is incomplete, containing records only up to 20 July 2026, and is treated accordingly wherever quarterly comparisons are made.

5. Dataset Description

The dataset used for this project contains Aadhaar enrollment records for the state of Maharashtra. It was loaded into the notebook as a CSV file using Pandas. The dataset consists of 537,489 rows and 7 original columns, with no missing values in any column.

5.1 Column Description

Column	Description	Data Type
date	Date of enrollment	Object (converted to datetime)
state	State name (Maharashtra for all records)	Object
district	District of enrollment	Object
pincode	Pincode of the enrollment location	Integer
age_0_5	Number of enrollments in the 0–5 years age group	Integer
age_5_17	Number of enrollments in the 5–17 years age group	Integer
age_18_greater	Number of enrollments aged 18 years and above	Integer

Table 5.1: Original dataset columns

5.2 Dataset Summary

- Total records: 537,489
- Number of unique districts: 36
- Number of unique pincodes: 1,604
- Date range: 1 January 2025 to 20 July 2026
- Missing values: None in any column
- Duplicate rows detected: 43,974 (identified during the data quality check, described in Section 8)

6. Tools and Technologies

Tool / Library	Purpose
Python	Core programming language used for the analysis
Pandas	Data loading, cleaning, grouping, and aggregation
NumPy	Numerical operations
Matplotlib	Line charts, bar charts, and scatter plots
Seaborn	Heatmap visualization
SciPy (scipy.stats)	Z-score calculation for anomaly detection
Jupyter Notebook (Google Colab)	Development environment for the analysis

Table 6.1: Tools and libraries used

7. Methodology

The project followed a structured data analysis workflow, moving from raw data to actionable insight in six stages:

- **Data Collection** — Loading the raw Aadhaar enrollment CSV file into a Pandas DataFrame.
- **Data Preparation** — Checking for missing values and duplicates, and converting the date column to a proper datetime format.
- **Feature Engineering** — Creating a total enrollment column and extracting year, month, month name, day, quarter, and year-month fields from the date.
- **Exploratory Data Analysis** — Studying enrollment patterns across time, district, pincode, and age group using summary statistics and visualizations.
- **Trend Identification** — Analyzing monthly, quarterly, and month-over-month growth trends, and applying z-score based anomaly detection.
- **Interpretation & Recommendations** — Converting the observed patterns into key findings and business recommendations.

Each stage is explained in detail in the following sections, in the order it was carried out in the notebook.

8. Data Cleaning and Preprocessing

8.1 Initial Inspection

The dataset was first inspected using `df.head()`, `df.tail()`, `df.info()`, `df.shape`, `df.columns.tolist()`, and `df.describe()` to understand its structure, size, and data types before any cleaning was performed.

8.2 Missing Value Check

`df.isnull().sum()` was used to check for missing values in every column. The check confirmed that none of the seven columns contained missing values.

8.3 Duplicate Check

`df.duplicated().sum()` was used to check for duplicate rows. This check returned 43,974 duplicate records in the dataset. These duplicates were identified as part of the data quality assessment; the notebook does not perform a removal step on them, and downstream aggregations in the analysis use the dataset as loaded.

8.4 Date Conversion

The date column was originally stored as an object (text) type in the format DD-MM-YYYY. It was converted to a proper datetime type using `pd.to_datetime()` with the format specified explicitly, which allows time-based operations such as extracting the year, month, and quarter later in the workflow.

8.5 Feature Engineering

A new column, `total_enrollment`, was created by summing the `age_0_5`, `age_5_17`, and `age_18_greater` columns for each row. This single column represents the total number of Aadhaar enrollments recorded at a given date and pincode, and is used throughout the rest of the analysis.

From the converted date column, five additional fields were derived:

- `year` — the calendar year of enrollment
- `month` — the numeric month of enrollment
- `month_name` — the full name of the month
- `day` — the day of the month
- `quarter` — the calendar quarter of enrollment

A combined `year_month` field (e.g. “2025-01”) was also created later in the notebook, used specifically for month-level grouping in the trend and anomaly analyses.

9. Exploratory Data Analysis

Before building any visualizations, basic exploratory checks were run on the dataset: the number of unique districts (36), the number of unique pincodes (1,604), the enrollment date range (1 January 2025 to 20 July 2026), summary statistics of total_enrollment (mean of 4.67 enrollments per record, maximum of 386), and the total enrollment split by age group (1,773,937 in the 0–5 years group, 676,739 in the 5–17 years group, and 57,266 in the 18+ years group).

9.1 Monthly Enrollment Trend

Purpose: To observe how total Aadhaar enrollment changed month by month across the study period.

Observation: A line chart of total enrollment against year-month shows a fluctuating pattern rather than a steady trend, with two clearly visible peaks.

Key Insight: Enrollment activity remained dynamic throughout the study period, with notable peaks during February and July 2025.

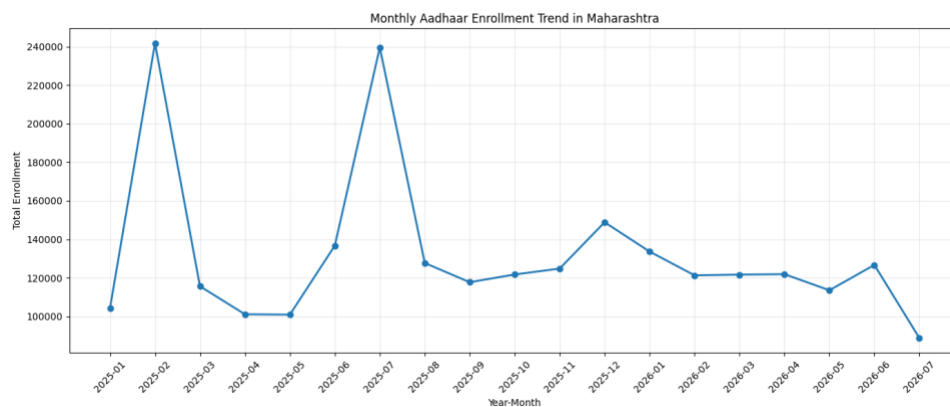


Figure 9.1 Monthly Aadhaar Enrollment Trend

9.2 District-wise Total Enrollment

Purpose: To compare cumulative Aadhaar enrollment across all 36 districts of Maharashtra.

Observation: A horizontal bar chart ranked districts by total enrollment. Pune recorded the highest total enrollment (233,827), followed by Thane (192,332), Mumbai Suburban (170,285), and Nashik (168,303).

Key Insight: Aadhaar enrollment shows significant geographical variation across Maharashtra, with Pune, Thane, Mumbai Suburban, and Nashik recording the highest enrollments, while several smaller districts show comparatively lower enrollment counts.

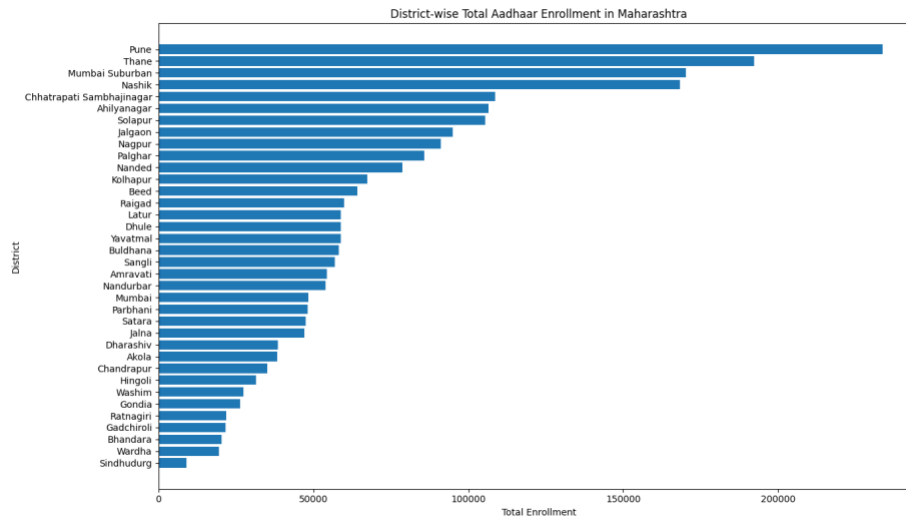


Figure 9.2 District-wise Total Aadhaar Enrollment

9.3 Age Group Distribution

Purpose: To understand how Aadhaar enrollment is distributed across the three defined age groups.

Observation: A bar chart of total enrollment by age group shows the 0–5 years group far ahead of the other two, followed by 5–17 years, with 18+ years recording the lowest count.

Key Insight: Aadhaar enrollment is predominantly concentrated in the 0–5 years age group, followed by 5–17 years, while the 18+ years category accounts for the smallest share of enrollments.

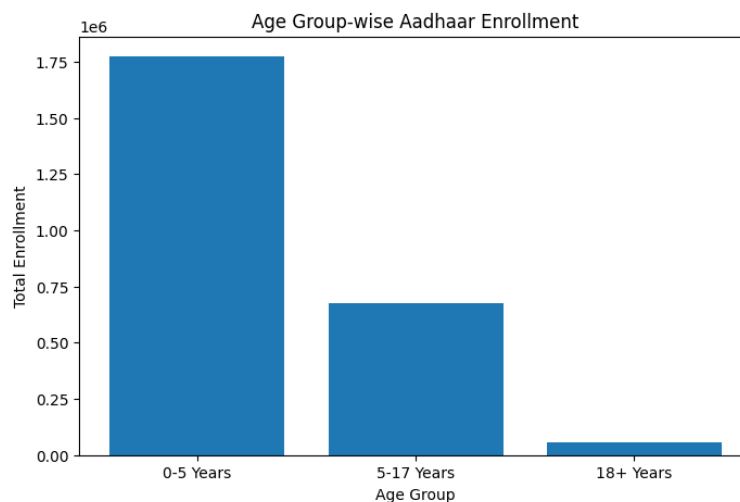


Figure 9.3 Age Group-wise Aadhaar Enrollment

9.4 Quarterly Enrollment Trend

Purpose: To study enrollment behaviour at a broader, quarterly level rather than month by month.

Observation: A line chart of total enrollment by year-quarter shows 2025-Q3 recording the highest enrollment (484,973) among all complete quarters, followed by 2025-Q1 (461,775). 2026-Q3 shows a much lower value, but it only contains data up to 20 July 2026 and is not a complete quarter.

Key Insight: Quarterly analysis highlights changing enrollment demand across the study period, with 2025-Q3 recording the highest overall enrollment among complete quarters.

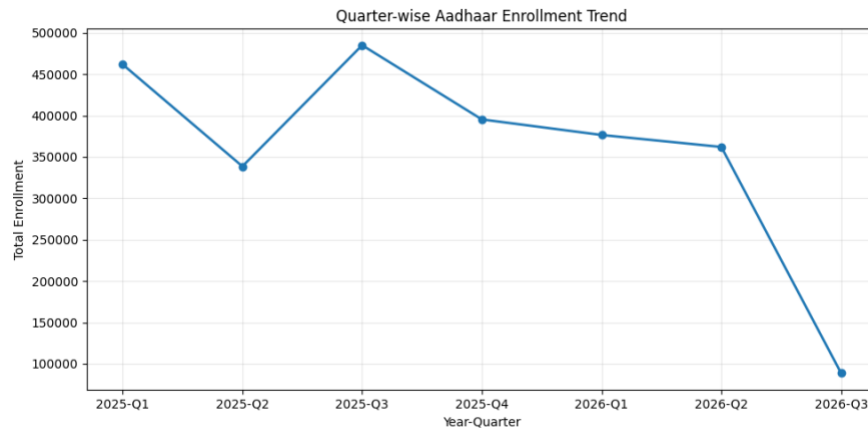


Figure 9.4 Quarter-wise Aadhaar Enrollment Trend

9.5 Top Pincodes by Aadhaar Enrollment

Purpose: To identify the specific pincodes contributing the most to total enrollment.

Observation: The top 10 pincodes by total enrollment were extracted and plotted. Pincode 431001 recorded the highest enrollment at 34,599, followed by 423203 (28,348) and 421302 (23,556).

Key Insight: A small number of pincodes contribute significantly to Aadhaar enrollment, indicating localized areas with consistently higher enrollment activity.

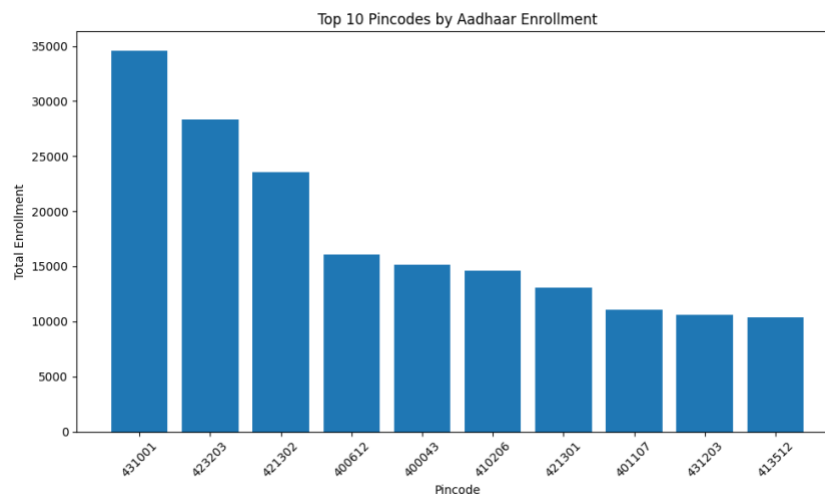


Figure 9.5 Top 10 Pincodes by Aadhaar Enrollment

9.6 Monthly Enrollment Heatmap by District

Purpose: To visualize enrollment intensity across both districts and months simultaneously.

Observation: A heatmap was built using district on one axis and year-month on the other, with cell values coloured by total enrollment using a YlGnBu colour scale.

Key Insight: Aadhaar enrollment intensity varies across both districts and months, highlighting districts with consistently high enrollment activity and periods of increased demand.

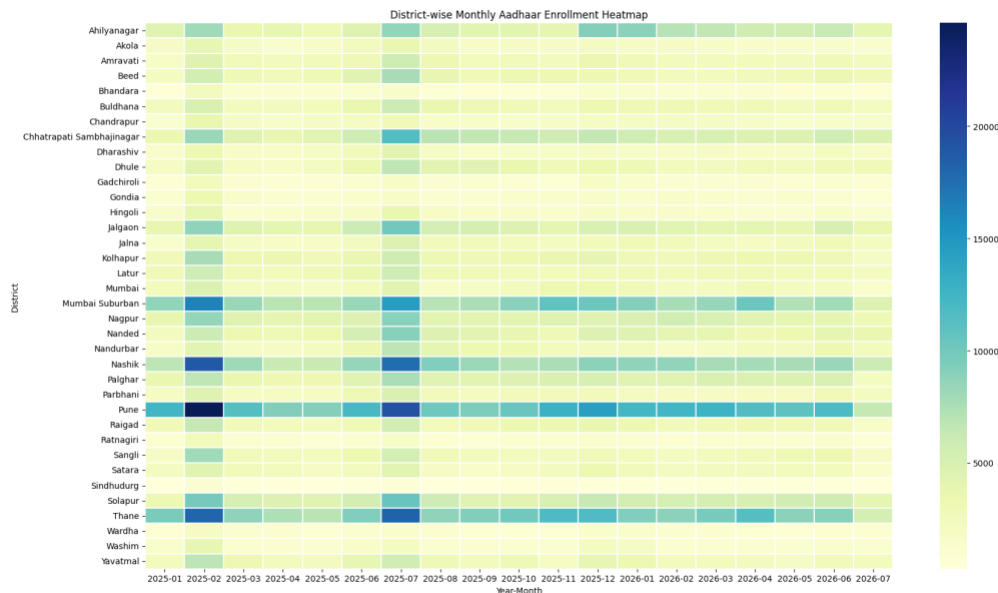


Figure 9.6 District-wise Monthly Enrollment Heatmap

9.7 Month-over-Month Growth Rate Analysis

Purpose: To measure how enrollment changed in percentage terms from one month to the next.

Observation: The percentage change in monthly total enrollment was calculated. February 2025 showed a sharp increase of 131.87% over January 2025, followed by a decline of -52.22% in March and -12.58% in April.

Key Insight: The month-over-month growth rate fluctuates throughout the study period, indicating periods of rapid enrollment growth as well as temporary declines.

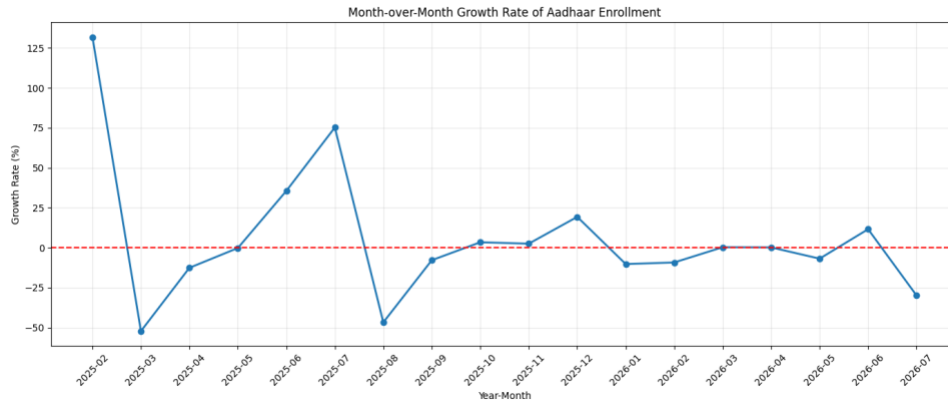


Figure 9.7 Month-over-Month Growth Rate

9.8 Top 10 Highest Enrollment Days

Purpose: To identify individual days with the highest recorded enrollment activity.

Observation: The top 10 days by total enrollment were extracted. 5 February 2025 recorded the highest single-day enrollment at 35,524, followed by 18 February 2025 (32,794) and 11 July 2025 (24,200).

Key Insight: A limited number of days recorded exceptionally high Aadhaar enrollment, indicating periods of concentrated enrollment activity that may correspond to special campaigns or increased public demand.

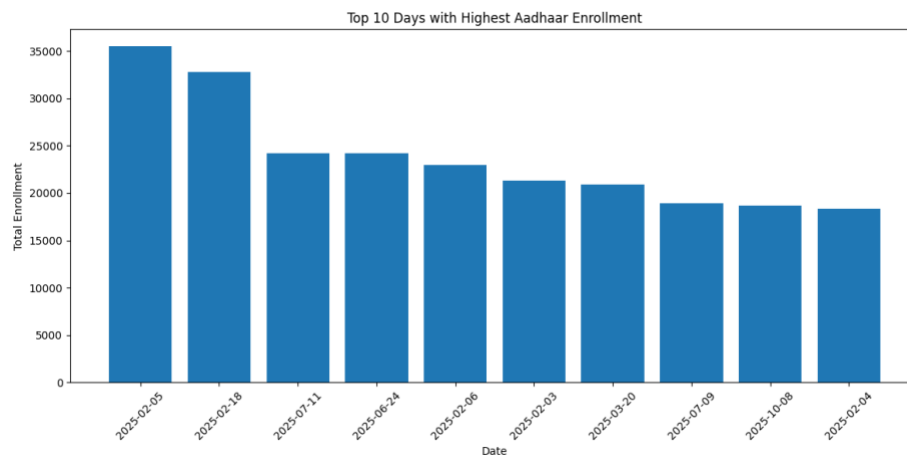


Figure 9.8 Top 10 Highest Enrollment Days

9.9 Anomaly Detection

Purpose: To statistically identify months where enrollment activity deviated significantly from the overall pattern.

Observation: A z-score was calculated for each month's total enrollment, and months with an absolute z-score greater than 2 were flagged as anomalies. February 2025 ($z = 2.77$) and July 2025 ($z = 2.71$) were identified as anomalous months.

Key Insight: The anomaly detection confirms that February and July 2025 were genuine outliers in enrollment activity rather than routine monthly variation, reinforcing the pattern seen in the monthly trend chart.

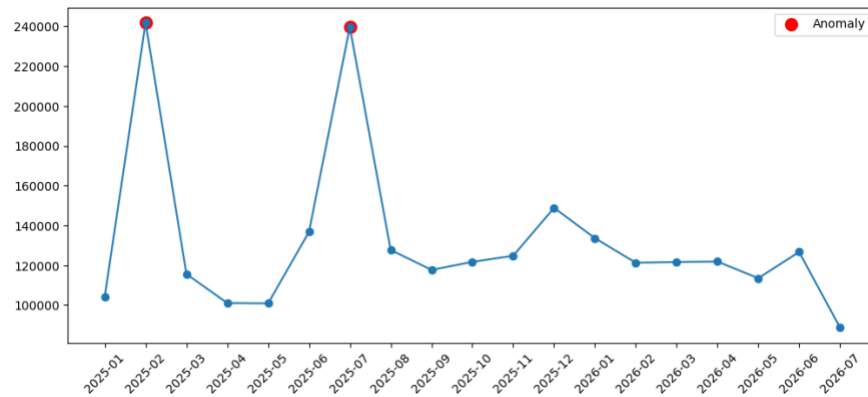


Figure 9.9 Monthly Enrollment with Anomalies Highlighted

10. Key Findings

- The dataset contains Aadhaar enrollment records for Maharashtra from January 2025 to July 2026, spanning 537,489 records across 36 districts and 1,604 pincodes.
- No missing values were found in the dataset; however, the duplicate check flagged 43,974 duplicate rows.
- Aadhaar enrollment varied across different months, with noticeable peaks during February 2025 and July 2025.
- Quarterly analysis showed fluctuations in enrollment, with 2025-Q3 recording the highest enrollment among the complete quarters analyzed.
- The 0–5 years age group accounted for the highest number of enrollments (1,773,937), followed by 5–17 years (676,739), while enrollments for individuals aged 18 years and above were comparatively lower (57,266).
- Enrollment activity varied considerably across districts, with Pune, Thane, Mumbai Suburban, and Nashik contributing the highest enrollment volumes.
- A small number of pincodes, led by 431001 and 423203, accounted for a significant share of total enrollments, indicating localized areas of high enrollment activity.
- The month-over-month growth analysis showed periods of both positive and negative growth, reflecting changing enrollment demand over time.
- Statistical anomaly detection using z-scores confirmed February and July 2025 as months with significantly elevated enrollment activity.

11. Challenges Faced

- Handling a large dataset of over 500,000 records while keeping grouping and aggregation operations efficient in Pandas.
- Deciding how to treat the duplicate rows identified during the data quality check, since the dataset spans daily enrollment entries where repeated pincode-level values across dates are expected to some extent.
- Interpreting the incomplete 2026-Q3 quarter correctly, since it only contains data up to 20 July 2026 and could otherwise be misread as a decline in enrollment.
- Choosing an appropriate statistical threshold (z-score greater than 2) for anomaly detection that would flag genuinely unusual months without over-flagging normal variation.
- Presenting district-wise and pincode-wise comparisons clearly given the large number of categories (36 districts and 1,604 pincodes) involved.

12. Conclusion

This project analyzed Aadhaar enrollment data for Maharashtra to identify temporal, geographical, and demographic enrollment patterns. Through data cleaning, feature engineering, and exploratory data analysis, several meaningful trends were identified, including monthly enrollment variations, district-level differences, pincode-wise distribution, and age-group participation.

The findings show how data analytics can support evidence-based planning and resource allocation for Aadhaar enrollment services. The methodology developed in this project can be extended to other states or larger datasets to enable broader policy analysis and decision-making.

12.1 Business Recommendations

- Increase enrollment resources and staffing during months with historically high enrollment activity.
- Conduct awareness campaigns in districts with comparatively lower enrollment to improve Aadhaar coverage.
- Monitor high-demand pincodes regularly to ensure adequate infrastructure and reduce waiting times.
- Develop targeted enrollment strategies for age groups with lower participation where appropriate.
- Continuously monitor monthly enrollment trends to identify unusual increases or decreases and investigate their causes.
- Implement periodic monitoring dashboards to continuously track enrollment trends across districts and age groups.

13. Future Scope

- Extend the analysis to include demographic and biometric update records, not only new enrollments.
- Compare Maharashtra's enrollment patterns with other states to identify relative performance.
- Build an interactive dashboard using a tool such as Power BI or Streamlit for real-time monitoring instead of static notebook charts.
- Investigate the root cause of the duplicate records identified during data cleaning and apply a de-duplication strategy before analysis.
- Apply forecasting models to predict enrollment volumes for upcoming months and support proactive resource planning.

14. References

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